Generic Number-of-Copies Amplification for Pseudorandom States

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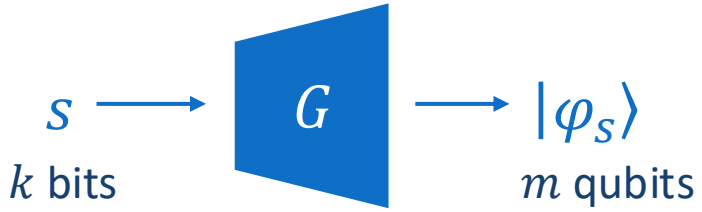
TCC 2026

Funded by the European Union

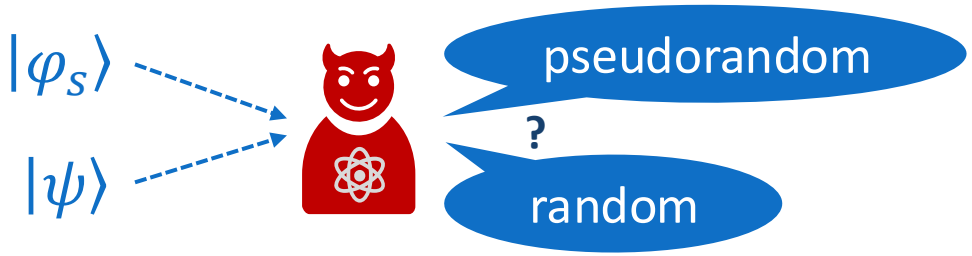
European Research Council
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Pseudorandom States [JLS18]

Efficiently generatable



Computationally indistinguishable from Haar random



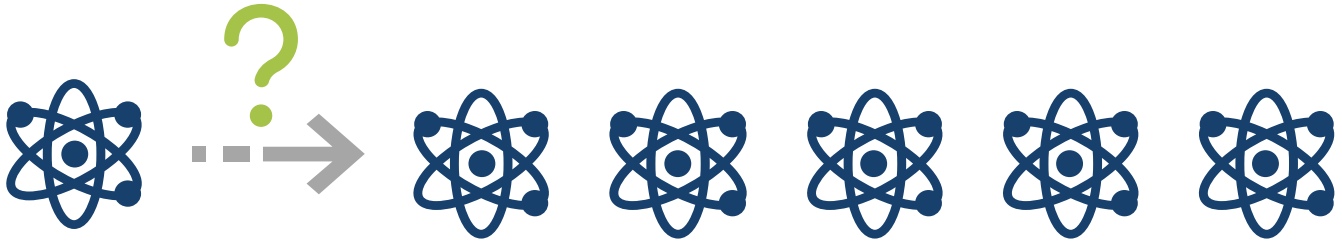
How many copies can the adversary use?

One Copy vs. Many Copies

Classical information



Quantum information



Central question:

Does security for fewer copies imply security for more copies?

» No-cloning: many copies can be a genuinely stronger resource

Motivation

- PRS may exist without one-way functions [Kretschmer21, KQST23]
- PRS enable cryptographic tasks:
  - **Commitment schemes** [AQY22, MY22]
  - One-time secure **digital signatures** with quantum public keys [MY22]
- Studying PRS helps in mapping out “Microcrypt”
  - Recently: 1-PRS $\Leftrightarrow$ EFI pairs [Bartra26]

Notions of Pseudorandom States

[JLS18, MY22, GJMZ23, BCN25, CCCGHJL25, CCS25, AG26, Bartra26…]

1-PRS
secure against 1 copy

t-PRS
secure against t copies

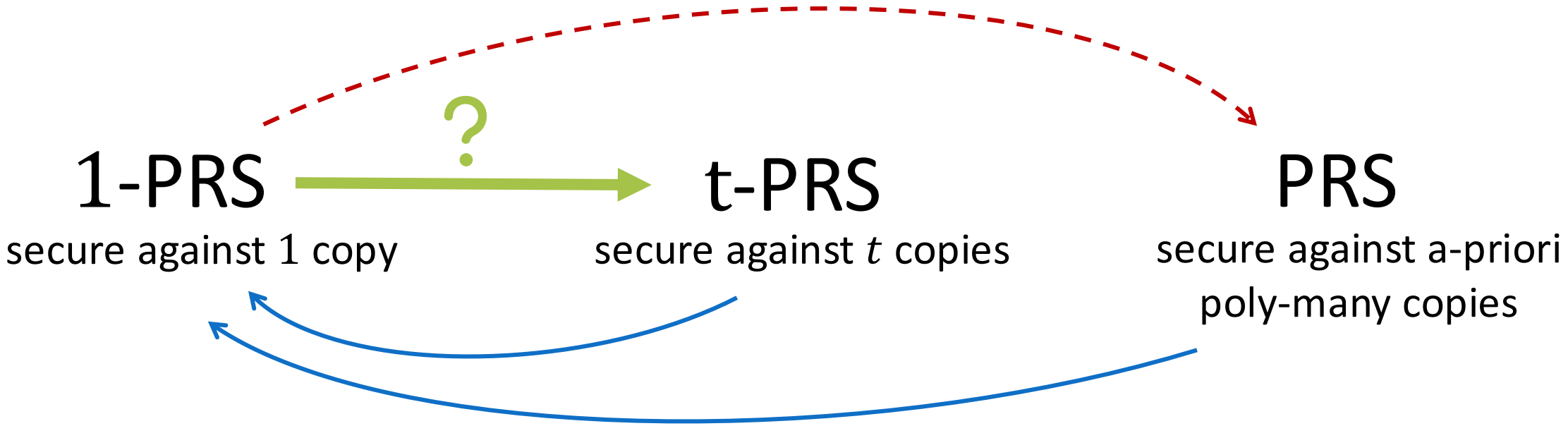
PRS
secure against a-priori poly-many copies

Non-trivial stretch: $s := tm - k > 0$

≫ For 1-PRS: $s := m - k > 0$

Notions of Pseudorandom States

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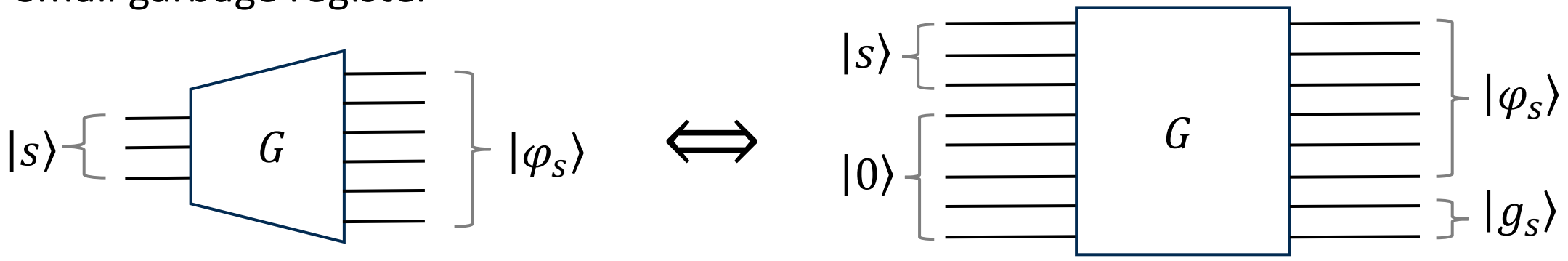
Prior State of the Art

Ananth-Goldin (CRYPTO 2026):

$$\forall t = \mathrm{poly}(\lambda) \quad \text{restricted } 1\text{−PRS} \implies t\text{−PRS}$$

Their restrictions:

- Large stretch, $m \gg k$
- Small garbage register



» **Our goal:** Remove both restrictions to achieve generic amplification.

Main Result

$\forall$ polynomial $t = t(\lambda)$

$$1\text{-PRS} \Rightarrow t\text{-PRS}$$

(no additional assumptions)

The AG Construction

High level outline:

1. Prepare a superposition of independently keyed 1-PRS states.

$$\frac{1}{2^{\lambda/2}} \sum_{i \in \{0,1\}^\lambda} |i\rangle$$

$$\downarrow$$

random function

$$\frac{1}{2^{\lambda/2}} \sum_{i \in \{0,1\}^\lambda} |i\rangle\,|f_2(i)\rangle|0 \dots 0\rangle$$

$$\downarrow \text{ apply } G$$

$$\frac{1}{2^{\lambda/2}} \sum_{i \in \{0,1\}^\lambda} |i\rangle\,|\varphi_{f_2(i)}\rangle|g_{f_2(i)}\rangle$$

The AG Construction

High level outline:

1. Prepare a superposition of independently keyed 1-PRS states.
2. Add random phases to eliminate unwanted correlations between the different terms.

$$\frac{1}{2^{\lambda/2}} \sum_{i \in \{0,1\}^{\lambda}} \omega_{t+1}^{f_1(i)} |i\rangle |\varphi_{f_2(i)}\rangle |g_{f_2(i)}\rangle$$

Random Phases

Consider the density matrix of t copies

t indices for the ket

$$\boldsymbol{i} = (i_1, \ldots, i_t)$$

t indices for the bra

$$\boldsymbol{i'} = (i'_1, \ldots, i'_t)$$

phase contributes

$$\mathbb{E}_{f_1}\left[\omega_{t+1}^{\sum_{j=1}^{t} f_1(i_j) - f_1(i'_j)}\right] = \begin{cases} 1, & \text{if } \boldsymbol{i'} = \boldsymbol{\pi}(\boldsymbol{i}) \\ 0, & \text{otherwise} \end{cases}$$

- Only premutation correlations survive!
- Similar structure to t copies of Haar state

The AG Construction

High level outline:

1. Prepare a superposition of independently keyed 1-PRS states.
2. Add random phases to eliminate unwanted correlations between the different terms.
3. Handle the garbage produced by coherent execution of G.

random functions

$$\frac{1}{2^{\lambda/2}} \sum_{i \in \{0,1\}^{\lambda}} \omega_{t+1}^{f_1(i)} |i\rangle |\varphi_{f_2(i)}\rangle |g_{f_2(i)}\rangle$$

(ignore garbage for now)

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Our improvement #1: binary phases suffice! Replace ω_{t+1} by $\omega_2 = -1$.

$\Longrightarrow$ simpler construction

Instantiating the Random Functions

$$\frac{1}{2^{\lambda/2}} \sum_{i \in \{0,1\}^{\lambda}} (-1)^{f_1(i)} |i\rangle |\varphi_{f_2(i)}\rangle |g_{f_2(i)}\rangle$$

$$\text{stretch} := t \cdot \text{output length} - \text{key length}$$

Thm [Zhandry12]: The random functions can be replaced by $2t$-wise independent functions.

Previous analysis [AG26]
f_1, f_2: $2t$-wise independent
$\text{stretch} = t(m - 2k - \lambda)$

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Thm [Zhandry12]: The random functions can be replaced by $2t$-wise independent functions.

Previous analysis [AG26]	Our analysis
f_1, f_2: $2t$-wise independent	f_1: $2t$-wise independent f_2: t-wise independent
$\text{stretch} = t(m - 2k - \lambda)$	$\text{stretch} = t(m - k - \lambda)$

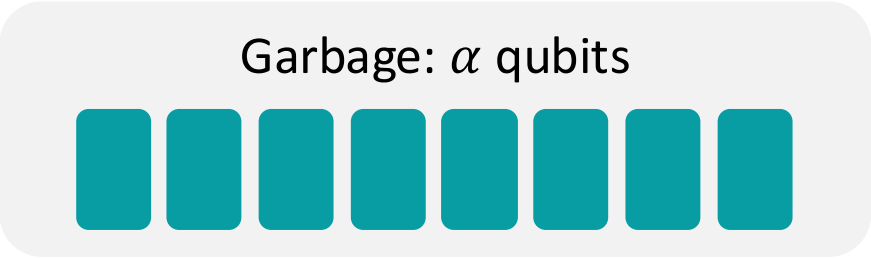
Our improvement #2: t-wise independence suffices for f_2!

$\Longrightarrow$ shorter description of f_2 $\Longrightarrow$ shorter key $\Longrightarrow$ larger stretch

The Garbage Bottleneck

$$\frac{1}{2^{\lambda/2}} \sum_{i \in \{0,1\}^{\lambda}} (-1)^{f_1(i)} |i\rangle |\varphi_{f_2(i)}\rangle |g_{f_2(i)}\rangle$$

Previous approach [AG26]



$\Downarrow$

QOTP on all α qubits

2α key bits

The Garbage Bottleneck

$$\frac{1}{2^{\lambda/2}} \sum_{i \in \{0,1\}^{\lambda}} (-1)^{f_1(i)} |i\rangle |\varphi_{f_2(i)}\rangle |g_{f_2(i)}\rangle$$

Previous approach [AG26]

Garbage: α qubits



⇓

QOTP on all α qubits

2α key bits

Our approach

inspired by the work of [CCCGHJL25]

Apply quantum extractor*



⇓

QOTP on roughly half

$\alpha + O(\lambda)$ key bits

*possible alternative: approximate quantum encryption [AS04]

Our improvement #3: Eliminate the garbage-size penalty in the stretch!

$$\Longrightarrow \text{stretch} = t\big(m - k - O(\lambda)\big)$$

Summary

Our improvement #1: binary phases

Our improvement #2: tighter independence analysis

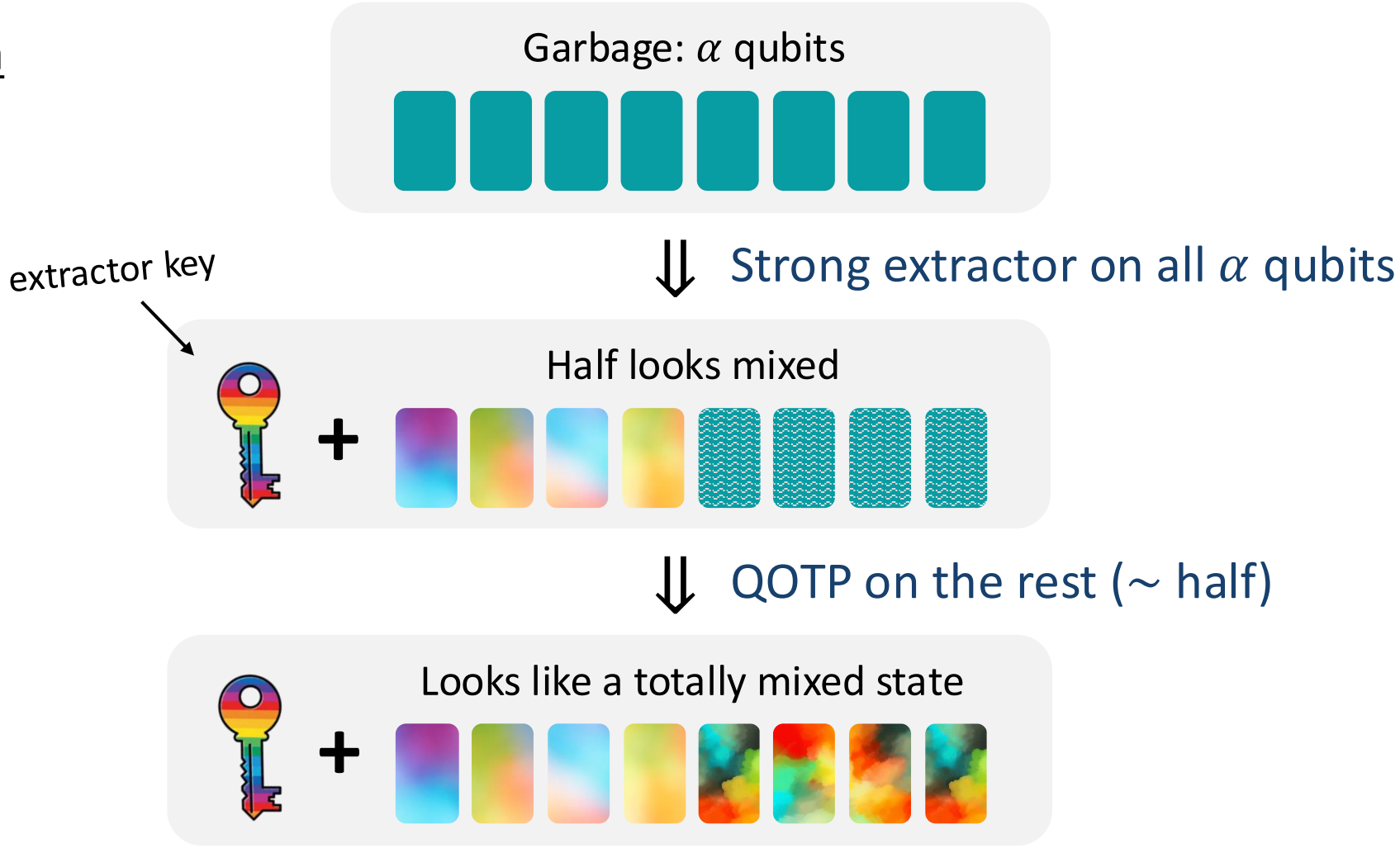
Our improvement #3: quantum extractor for the garbage

Main result: $\forall t = \mathrm{poly}(\lambda) \quad 1\text{−PRS} \implies t\text{−PRS}$

https://arxiv.org/abs/2606.29325

The Garbage Bottleneck

Our approach



Complete Security Hybrid Proof

H_0- Actual construction, t copies

↓ distinct indices; random phases enforce permutation structure

H_1- Symmetrically decoupled state

↓ extractor + QOTP

H_2- Garbage and extractor-seed registers look uniform

↓ use 1-PRS security

H_3- Replace the 1-PRS outputs by maximally mixed states

↓ symmetric-subspace argument

H_4- t copies of a Haar-random state